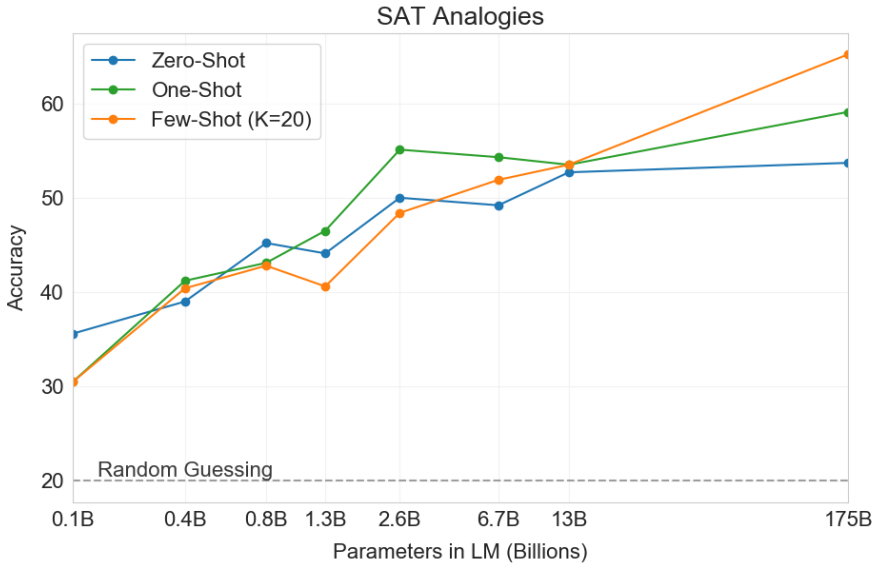

Figure 3.12: Zero-, one-,and few-shot performance on SAT analogy tasks, for different sizes of model. The largest model achieves 65% accuracy in the few-shot setting, and also demonstrates significant gains to in-context learning which are not present in smaller models.

3.9.4 News Article Generation

Previous work on generative language models qualitatively tested their ability to generate synthetic "news articles" by conditional sampling from the model given a human-written prompt consisting of a plausible first sentence for a news story [RWC+19]. Relative to [RWC+19], the dataset used to train GPT-3 is much less weighted towards news articles, so trying to generate news articles via raw unconditional samples is less effective – for example GPT-3 often interprets the proposed first sentence of a "news article" as a tweet and then posts synthetic responses or follow-up tweets. To solve this problem we employed GPT-3's few-shot learning abilities by providing three previous news articles in the model's context to condition it. With the title and subtitle of a proposed next article, the model is able to reliably generate short articles in the "news" genre.

To gauge the quality of news article generation from GPT-3 (which we believe is likely to be correlated with conditional sample generation quality in general), we decided to measure human ability to distinguish GPT-3-generated articles from real ones. Similar work has been carried out by Kreps et al. [KMB20] and Zellers et al. [ZHR+19]. Generative language models are trained to match the distribution of content generated by humans, so the (in)ability of humans to distinguish the two is a potentially important measure of quality.[3]

In order to see how well humans can detect model generated text, we arbitrarily selected 25 article titles and subtitles from the website newser.com (mean length: 215 words). We then generated completions of these titles and subtitles from four language models ranging in size from 125M to 175B (GPT-3) parameters (mean length: 200 words). For each model, we presented around 80 US-based participants with a quiz consisting of these real titles and subtitles followed by either the human written article or the article generated by the model[4]. Participants were asked to select whether the article was "very likely written by a human", "more likely written by a human", "I don't know", "more likely written by a machine", or "very likely written by a machine".

The articles we selected were not in the models' training data and the model outputs were formatted and selected programmatically to prevent human cherry-picking. All models used the same context to condition outputs on and were pre-trained with the same context size and the same article titles and subtitles were used as prompts for each model. However, we also ran an experiment to control for participant effort and attention that followed the same format but involved intentionally bad model generated articles. This was done by generating articles from a "control model": a 160M parameter model with no context and increased output randomness.

[3]This task is also relevant to the potential misuse of language models discussed in Section 6.1.

[4]We wanted to identify how good an average person on the internet is at detecting language model outputs, so we focused on participants drawn from the general US population. See Appendix E for details.

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p-value)	"I don't know" assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 ($2e$-4)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 ($7e$-21)	6.0%
GPT-3 Large	68%	64%–72%	7.3 ($3e$-11)	8.7%
GPT-3 XL	62%	59%–65%	10.7 ($1e$-19)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 ($5e$-19)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 ($3e$-21)	6.2%
GPT-3 13B	55%	52%–58%	15.3 ($1e$-32)	7.1%
GPT-3 175B	52%	49%–54%	16.9 ($1e$-34)	7.8%

Table 3.11: Human accuracy in identifying whether short ($\sim$200 word) news articles are model generated. We find that human accuracy (measured by the ratio of correct assignments to non-neutral assignments) ranges from 86% on the control model to 52% on GPT-3 175B. This table compares mean accuracy between five different models, and shows the results of a two-sample T-Test for the difference in mean accuracy between each model and the control model (an unconditional GPT-3 Small model with increased output randomness).

Mean human accuracy (the ratio of correct assignments to non-neutral assignments per participant) at detecting that the intentionally bad articles were model generated was $\sim 86\%$ where 50% is chance level performance. By contrast, mean human accuracy at detecting articles that were produced by the 175B parameter model was barely above chance at $\sim 52\%$ (see Table 3.11).[5] Human abilities to detect model generated text appear to decrease as model size increases: there appears to be a trend towards chance accuracy with model size, and human detection of GPT-3 is close to chance.[6] This is true despite the fact that participants spend more time on each output as model size increases (see Appendix E).

Examples of synthetic articles from GPT-3 are given in Figures 3.14 and 3.15.[7] Much of the text is—as indicated by the evaluations—difficult for humans to distinguish from authentic human content. Factual inaccuracies can be an indicator that an article is model generated since, unlike human authors, the models have no access to the specific facts that the article titles refer to or when the article was written. Other indicators include repetition, non sequiturs, and unusual phrasings, though these are often subtle enough that they are not noticed.

Related work on language model detection by Ippolito et al. [IDCBE19] indicates that automatic discriminators like GROVER [ZHR+19] and GLTR [GSR19] may have greater success at detecting model generated text than human evaluators. Automatic detection of these models may be a promising area of future research.

Ippolito et al. [IDCBE19] also note that human accuracy at detecting model generated text increases as humans observe more tokens. To do a preliminary investigation of how good humans are at detecting longer news articles generated by GPT-3 175B, we selected 12 world news articles from Reuters with an average length of 569 words and generated completions of these articles from GPT-3 with an average length of 498 words (298 words longer than our initial experiments). Following the methodology above, we ran two experiments, each on around 80 US-based participants, to compare human abilities to detect the articles generated by GPT-3 and a control model.

We found that mean human accuracy at detecting the intentionally bad longer articles from the control model was $\sim 88\%$, while mean human accuracy at detecting the longer articles that were produced by GPT-3 175B was still barely above chance at $\sim 52\%$ (see Table 3.12). This indicates that, for news articles that are around 500 words long, GPT-3 continues to produce articles that humans find difficult to distinguish from human written news articles.

3.9.5 Learning and Using Novel Words

A task studied in developmental linguistics [CB78] is the ability to learn and utilize new words, for example using a word in a sentence after seeing it defined only once, or conversely inferring a word's meaning from only one usage. Here we qualitatively test GPT-3's ability to do the former. Specifically, we give GPT-3 the definition of a nonexistent word, such as "Gigamuru", and then ask it to use it in a sentence. We provide one to five previous examples of a (separate)

[5] We use a two-sample Student's T-Test to test for significant difference between the means of the participant accuracies of each model and the control model and report the normalized difference in the means (as the t-statistic) and the p-value.

[6] If a model consistently produces texts that are more impressive than human articles, it is possible that human performance on this task would drop below 50%. Indeed, many individual participants scored below 50% on this task.

[7] Additional non-news samples can be found in Appendix F.

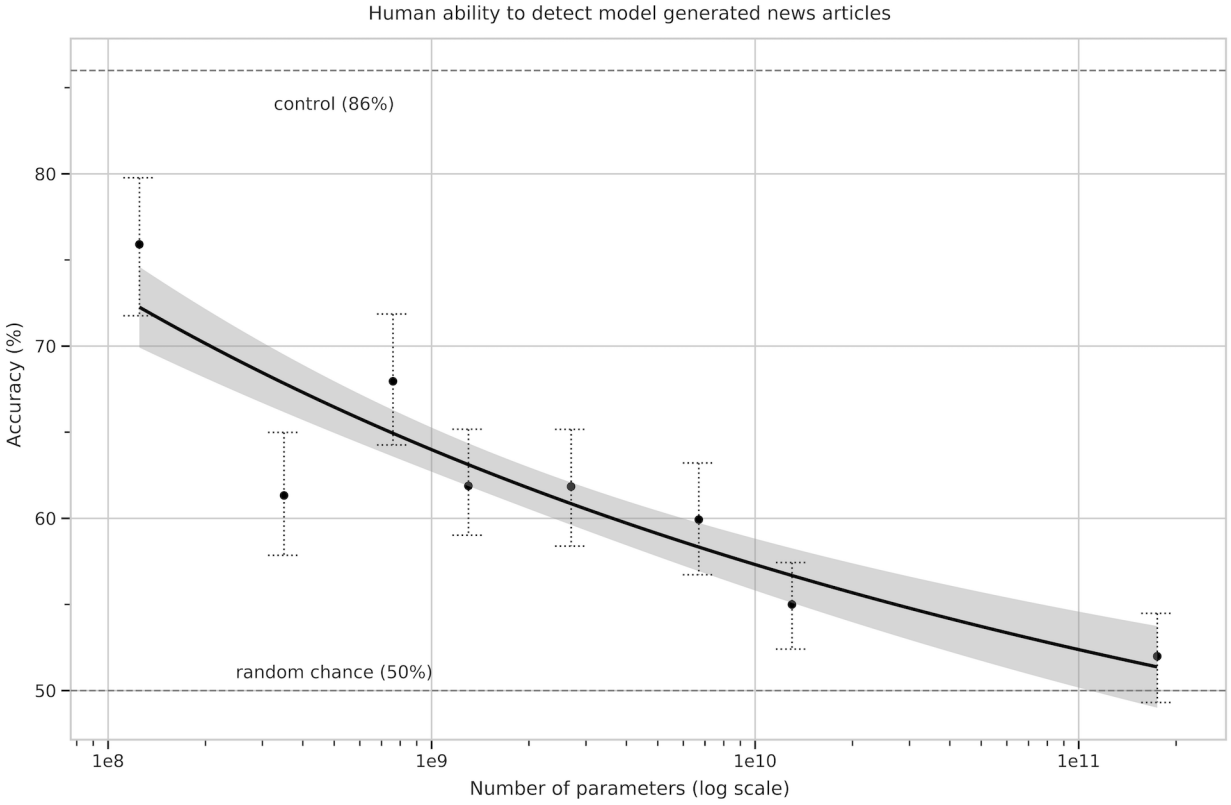

Figure 3.13: People's ability to identify whether news articles are model-generated (measured by the ratio of correct assignments to non-neutral assignments) decreases as model size increases. Accuracy on the outputs on the deliberately-bad control model (an unconditioned GPT-3 Small model with higher output randomness) is indicated with the dashed line at the top, and the random chance (50%) is indicated with the dashed line at the bottom. Line of best fit is a power law with 95% confidence intervals.

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p-value)	"I don't know" assignments
Control	88%	84%–91%	-	2.7%
GPT-3 175B	52%	48%–57%	12.7 (3.2e-23)	10.6%

Table 3.12: People's ability to identify whether ~ 500 word articles are model generated (as measured by the ratio of correct assignments to non-neutral assignments) was 88% on the control model and 52% on GPT-3 175B. This table shows the results of a two-sample T-Test for the difference in mean accuracy between GPT-3 175B and the control model (an unconditional GPT-3 Small model with increased output randomness).